

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 20

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Negative feedback responds to any change by re-establishing the previous equilibrium • Only large unusual events overwhelm the feedback so a new equilibrium forms • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction	3
01.2	• In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source	4
01.3	<i>Any 3 from:</i> • Melting permafrost releasing methane • Reduced albedo as ice / snow melt • Methane hydrate release • Forest and peat fires • Faster soil decomposition • Ocean acidification reducing sequestration • Cirrus cloud formation	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	

Question	Marking guidance	Marks
02.1	• In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching)	2
02.2	<i>Any 4 from:</i> • Is the wild population threatened? • Is there a genetically diverse captive population? • Is in-situ conservation failing? • Is keeping a captive population realistic (habitat size, food, relationships, cost)? • Is release likely to be successful now or in future (suitable habitat, threats removed)?	4
02.3	• Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
03.1	• $600 + (90 + 150) - (220 + 80) = 540$ • Stock fell by 60 so the catch exceeds the MSY / unsustainable	2
03.2	• $A + G - M = 90 + 150 - 80 = 160$	1
03.3	• Catch is landed and weighed at ports so it is recorded • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years	3
03.4	• CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
04.1	• Isolation makes colonisation difficult: only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them • Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists • Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) • Endemics have small ranges and populations so are vulnerable to extinction	4

Question	Marking guidance	Marks
04.2	• The birds evolved without mammal predators so nest on the ground / in burrows with no defence • Rats eat eggs and chicks; numbers fell to very low levels (45 and 12 pairs) • After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor	3
04.3	• Introduced mosquitoes carry avian malaria to which endemic birds have no resistance; birds survive only above about 1500 m where it is too cold for malaria • Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range	2
04.4	• Control of developments and visitor numbers (biosecurity, quarantine of imports); protected areas; captive breeding of endemics	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	
05.1	• Success: Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia; Encarsia parasitic wasps control greenhouse whitefly • Problems: cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators; rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct); harlequin ladybirds introduced to Europe eat native ladybirds • Lesson: control agents should be specialist feeders on the pest	4
05.2	• Release an artificial sex scent attracting males (or females) of the pest species • Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding	2
05.3	• Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply • Agricultural use exceeds medical use, so resistance spreads • Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat; hence reduced use of antibiotics is a sustainability goal	3
05.4	• Grow nectar-rich flowering plants alongside crops; restrict pesticides harmful to bees / spray when not in flower; introduce bee hives	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
06.1	<i>Any 4 from:</i> • Copying the structural / functional adaptations of living organisms in the design of manufactured items • Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use • Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) • Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) • Termite mound convection copied for natural ventilation of buildings; hollow strutted bird bones in lightweight roofs; gecko toe adhesion; burr hooks - Velcro; lotus leaf self-cleaning glass; spider silk for airbags / body armour; bat / dolphin echolocation - ultrasound scanners	4
06.2	• Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores; in controlled doses these affect human physiology • Examples: morphine / codeine from poppies; quinine from cinchona bark (malaria); aspirin from willow bark; taxol from yew (cancers); diosgenin from Mexican yam (contraceptives, cortisone) • Animal examples accepted: AZT from a marine sponge (HIV), shark liver lipids (tumours)	3
06.3	• Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals • Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships)	2
06.4	• Squid giant nerve cells - nerve function / heart disease, stroke; marsupials (kangaroo, wallaby pouch) - embryo development; hippo skin secretion - sunscreen / burns; sea urchin embryos - teratogen testing; armadillo - leprosy; marine sponge proteins - transplant rejection	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	
07.1	• Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance; indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met • Single species: easy planting, thinning and felling over large areas; but each tree species supports its own community so few species overall • Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them • Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind; no old trees or dead wood for invertebrates and woodpeckers • Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species	6
07.2	• Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline • Continuing to grow them ties up land and capital for little extra timber; natural forests keep old and dead trees	2
07.3	• Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp • Wood fuel replaced by fossil fuels / electricity; so traditional management declined and woods became uniform	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
08.1	• Food: cod, tuna, swordfish, sharks, turtles; extinct - dodo, great auk, passenger pigeon • Fashion: leopard, snow leopard, tiger, ocelot fur; fur seals; crocodile / alligator leather; kingfisher, parrot, ostrich feathers • Entertainment / pets: parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants • Furniture / ornaments: mahogany, teak, ramin timber; ebony and ivory piano keys; turtle shell, shark teeth, coral and shells as souvenirs • Traditional medicines: tiger (bones, claws), rhinoceros horn, seahorses	5
08.2	<i>Any 3 from:</i> • Threat to humans: sharks, venomous snakes, crocodiles • Disease vectors: malaria mosquitoes, tsetse flies • Predators of livestock: wolves, puma, birds of prey, herons, seals • Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers	3
08.3	• High demand and prices make illegal trade / poaching profitable; laws are hard to enforce across borders • Belief in the medicine persists despite little scientific evidence; species become rarer and more valuable	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	
09.1	<i>Any 2 from:</i> • Hydrological cycle • Carbon cycle • Nitrogen cycle • Atmospheric circulation • Thermohaline circulation • Photosynthesis	2
09.2	• Enzymes reduce the activation energy of reactions • So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures	2
09.3	<i>Any 1 from:</i> • Haber process (nitrogen fixation) • Incineration of wastes (decomposition) • Many manufacturing processes	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
10.2	• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25

Question	Marking guidance	Marks
11.2	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25